

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 4: Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Word problems

1. Over a 200 km journey, a fast train takes one hour less than a slow train. The slow train runs 10 km/h slower than the fast one. Find the speed of each train. **CBSE 2026, 5 marks**

SOLUTION

Let the fast train run at x km/h. Then the slow train runs at $(x - 10)$ km/h.

$$\frac{200}{x - 10} - \frac{200}{x} = 1$$

$$200x - 200(x - 10) = x(x - 10), \text{ so } 2000 = x^2 - 10x$$

$$x^2 - 10x - 2000 = 0, \text{ so } (x - 50)(x + 40) = 0$$

$x = 50$ or $x = -40$. A speed cannot be negative.

Check: $200 \div 50 = 4$ hours and $200 \div 40 = 5$ hours, which is one hour more.

Answer: 50 km/h and 40 km/h

2. Two squares have a total area of 640 m^2 , and their perimeters differ by 64 m. Find the side of each square. **CBSE 2026, 5 marks**

SOLUTION

Let the sides be x m and y m with $x > y$.

$$4x - 4y = 64, \text{ so } x = y + 16.$$

$$x^2 + y^2 = 640: (y + 16)^2 + y^2 = 640, \text{ so } 2y^2 + 32y - 384 = 0$$

$$y^2 + 16y - 192 = 0, \text{ so } (y + 24)(y - 8) = 0$$

$y = 8$ (a length cannot be negative) and $x = 24$.

Check: $24^2 + 8^2 = 576 + 64 = 640$.

Answer: 24 m and 8 m

3. A right-angled triangle has a perimeter of 60 cm and a hypotenuse of 25 cm. Find the lengths of the other two sides. **CBSE 2025, 5 marks**

SOLUTION

Let one of the other sides be x cm. The two sides add up to $60 - 25 = 35$ cm, so the other side is $(35 - x)$ cm.

By the Pythagoras theorem, $x^2 + (35 - x)^2 = 25^2$.

$x^2 + 1225 - 70x + x^2 = 625$, so $2x^2 - 70x + 600 = 0$, that is $x^2 - 35x + 300 = 0$.

$(x - 15)(x - 20) = 0$, so $x = 15$ or $x = 20$.

If $x = 15$ the other side is 20, and if $x = 20$ the other side is 15.

Answer: 15 cm and 20 cm

4. A train covers 480 km at a uniform speed. Had it travelled 8 km/h slower, the journey would have taken 3 hours longer. Find the speed of the train. **CBSE 2025, 5 marks**

SOLUTION

Let the speed be x km/h. The usual time is $\frac{480}{x}$ hours and the slower time is $\frac{480}{x - 8}$ hours.

$$\frac{480}{x - 8} - \frac{480}{x} = 3$$

$480x - 480(x - 8) = 3x(x - 8)$, so $3840 = 3x^2 - 24x$

$x^2 - 8x - 1280 = 0$, so $(x - 40)(x + 32) = 0$

$x = 40$ or $x = -32$. A speed cannot be negative.

Check: $480 \div 40 = 12$ hours and $480 \div 32 = 15$ hours, which is 3 hours more.

Answer: 40 km/h

5. A rectangular floor is covered exactly by 200 square tiles. With tiles whose side is 1 unit longer, only 128 tiles are needed. Let the side of an original tile be x units. (i) Form a quadratic equation in x . (ii) Write it in standard form. (iii) Find x by factorisation, OR solve the equation by the quadratic formula.

CBSE 2024, 4 marks, case study

SOLUTION

(i) The floor area is the same in both cases: $200x^2 = 128(x + 1)^2$

(ii) $200x^2 = 128x^2 + 256x + 128$, so $72x^2 - 256x - 128 = 0$. Dividing by 8: $9x^2 - 32x - 16 = 0$

(iii) By factorisation: $9x^2 - 36x + 4x - 16 = 0$, so $9x(x - 4) + 4(x - 4) = 0$ and $(9x + 4)(x - 4) = 0$

$x = 4$ or $x = -\frac{4}{9}$. A length cannot be negative, so $x = 4$.

OR by the formula: $x = \frac{32 \pm \sqrt{1024 + 576}}{18} = \frac{32 \pm 40}{18}$, which gives $x = 4$ or $x = -\frac{4}{9}$.

Answer: the side of a tile is 4 units.

6. A school photo is 18 cm long and 12 cm wide. Its length and width are each to be increased by x cm so that the area doubles. (i) Write an equation for this. (ii) Write it as a quadratic equation in standard form. (iii) Find the new length and width, OR decide whether any rational value of x gives a new area of 220 cm². **CBSE 2023, 4 marks**

SOLUTION

(i) Original area = $18 \times 12 = 216$ cm², so the new area is 432 cm². $(18 + x)(12 + x) = 432$

(ii) $216 + 30x + x^2 = 432$. $x^2 + 30x - 216 = 0$

(iii) $(x + 36)(x - 6) = 0$, so $x = 6$ (a length cannot be negative). **New size: 24 cm by 18 cm.**

OR: $(18 + x)(12 + x) = 220$ gives $x^2 + 30x - 4 = 0$, with $D = 900 + 16 = 916$. This is not a perfect square, so the roots are irrational. **No rational value of x works.**

7. Rehan's age 5 years ago multiplied by his age 7 years from now is one more than twice his present age. Find his present age. **CBSE 2022 Term 2, 2 marks**

SOLUTION

Let his present age be x years. Then $(x - 5)(x + 7) = 2x + 1$.

$x^2 + 2x - 35 = 2x + 1$, so $x^2 = 36$ and $x = \pm 6$.

An age cannot be negative.

Answer: 6 years

8. Two taps together fill a tank in $1\frac{7}{8}$ hours. Working alone, the larger tap takes 2 hours less than the smaller one. How long does each tap take on its own? **CBSE 2019, 4 marks**

SOLUTION

Let the larger tap take x hours. Then the smaller tap takes $x + 2$ hours, and together they take $\frac{15}{8}$ hours.

In one hour: $\frac{1}{x} + \frac{1}{x+2} = \frac{8}{15}$

$15(2x + 2) = 8x(x + 2)$, so $8x^2 - 14x - 30 = 0$, that is $4x^2 - 7x - 15 = 0$

$(4x + 5)(x - 3) = 0$, so $x = 3$ (time cannot be negative).

Answer: the larger tap takes 3 hours and the smaller tap takes 5 hours

9. A plane took off 30 minutes late. To reach its destination 1500 km away on time, it flew 100 km/h faster than usual. Find its usual speed. **CBSE 2018, 3 marks**

SOLUTION

Let the usual speed be x km/h. The faster flight saves half an hour.

$\frac{1500}{x} - \frac{1500}{x+100} = \frac{1}{2}$

$1500 \times 100 \times 2 = x(x + 100)$, so $x^2 + 100x - 300000 = 0$

$(x + 600)(x - 500) = 0$, so $x = 500$ (speed cannot be negative).

Answer: 500 km/h

- 10.** A motor boat has a speed of 18 km/h in still water. It takes 1 hour longer to go 24 km upstream than to come back the same distance downstream. Find the speed of the stream. **CBSE 2018, 4 marks**

SOLUTION

Let the speed of the stream be x km/h. Upstream speed = $18 - x$ and downstream speed = $18 + x$.

$$\frac{24}{18 - x} - \frac{24}{18 + x} = 1$$

$$24(18 + x) - 24(18 - x) = 324 - x^2, \text{ so } 48x = 324 - x^2 \text{ and } x^2 + 48x - 324 = 0$$

$$(x + 54)(x - 6) = 0, \text{ so } x = 6.$$

Answer: 6 km/h

- 11.** A train covers 63 km at a certain average speed and then 72 km at a speed 6 km/h faster. The whole journey takes 3 hours. Find the first speed. **CBSE 2018, 4 marks**

SOLUTION

$$\text{Let the first speed be } x \text{ km/h. } \frac{63}{x} + \frac{72}{x + 6} = 3$$

$$63(x + 6) + 72x = 3x(x + 6), \text{ so } 135x + 378 = 3x^2 + 18x$$

$$3x^2 - 117x - 378 = 0, \text{ that is } x^2 - 39x - 126 = 0$$

$$(x - 42)(x + 3) = 0, \text{ so } x = 42.$$

Answer: 42 km/h

- 12.** A motor boat has a speed of 24 km/h in still water. It takes 1 hour longer to go 32 km upstream than to return the same distance downstream. Find the speed of the stream. **CBSE 2016, 4 marks**

SOLUTION

Let the speed of the stream be x km/h.

$$\frac{32}{24 - x} - \frac{32}{24 + x} = 1$$

$$32(24 + x) - 32(24 - x) = 576 - x^2, \text{ so } 64x = 576 - x^2 \text{ and } x^2 + 64x - 576 = 0$$

$$(x + 72)(x - 8) = 0, \text{ so } x = 8.$$

Answer: 8 km/h

2 Nature of roots

- 13.** The equation $ax^2 + bx + c = 0$, $a \neq 0$, has real and equal roots. Which relation holds? (a) $a = \frac{b^2}{c}$ (b) $b^2 = ac$ (c) $ac = \frac{b^2}{4}$ (d) $c = \frac{b^2}{a}$ **CBSE 2024, 1 mark**

SOLUTION

For real and equal roots, $D = b^2 - 4ac = 0$.

$$\text{So } b^2 = 4ac, \text{ that is } ac = \frac{b^2}{4}.$$

Answer: (c)

- 14.** What is the least positive value of k for which $2x^2 + kx - 4 = 0$ has rational roots? (a) $\pm 2\sqrt{2}$ (b) 2 (c) ± 2 (d) $\sqrt{2}$ **CBSE 2023, 1 mark**

SOLUTION

The roots are rational when the discriminant is a perfect square. $D = k^2 - 4(2)(-4) = k^2 + 32$.

Among the options, $k = \sqrt{2}$ gives $D = 34$, which is not a perfect square. $k = 2$ gives $D = 36 = 6^2$.

Answer: (b)

- 15.** For what value of m does $(m - 1)x^2 + 2(m - 1)x + 1 = 0$ have two equal real roots?

CBSE 2022 Term 2, 2 marks

SOLUTION

Equal roots need $D = 0$: $[2(m - 1)]^2 - 4(m - 1)(1) = 0$

$4(m - 1)(m - 1 - 1) = 0$, so $m = 1$ or $m = 2$.

If $m = 1$ the x^2 term disappears and the equation is no longer quadratic, so $m = 1$ is rejected.

Answer: $m = 2$

- 16.** For what values of k does $x^2 + 4x + k = 0$ have real roots? **CBSE 2019, 1 mark**

SOLUTION

Real roots need $D \geq 0$: $16 - 4k \geq 0$

Answer: $k \leq 4$

- 17.** The roots of $3x^2 - 10x + k = 0$ are reciprocals of each other. Find k . **CBSE 2019, 1 mark**

SOLUTION

If the roots are α and $\frac{1}{\alpha}$, their product is 1.

Product of roots = $\frac{k}{3} = 1$

Answer: $k = 3$

- 18.** -5 is a root of $2x^2 + px - 15 = 0$, and the equation $p(x^2 + x) + k = 0$ has equal roots. Find k .

CBSE 2016, 2 marks

SOLUTION

Put $x = -5$ in the first equation: $50 - 5p - 15 = 0$, so $p = 7$.

The second equation is $7x^2 + 7x + k = 0$. Equal roots need $D = 0$: $49 - 28k = 0$.

Answer: $k = \frac{7}{4}$

3 Solving by factorisation

19. Solve for x : $\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$. **CBSE 2022 Term 2, 2 marks**

SOLUTION

Product of the first and last coefficients $= \sqrt{3} \times 7\sqrt{3} = 21 = 3 \times 7$, and $3 + 7 = 10$.

$$\sqrt{3}x^2 + 3x + 7x + 7\sqrt{3} = 0, \text{ so } \sqrt{3}x(x + \sqrt{3}) + 7(x + \sqrt{3}) = 0$$

$$(x + \sqrt{3})(\sqrt{3}x + 7) = 0$$

$$\text{Answer: } x = -\sqrt{3} \text{ or } x = -\frac{7}{\sqrt{3}} = -\frac{7\sqrt{3}}{3}$$

20. $x = 3$ is a root of $x^2 - 2kx - 6 = 0$. Find k . **CBSE 2018, 1 mark**

SOLUTION

Put $x = 3$: $9 - 6k - 6 = 0$, so $6k = 3$.

$$\text{Answer: } k = \frac{1}{2}$$

21. Solve for x : $\frac{1}{(x-1)(x-2)} + \frac{1}{(x-2)(x-3)} = \frac{2}{3}$, where $x \neq 1, 2, 3$. **CBSE 2016, 3 marks**

SOLUTION

$$\text{LHS} = \frac{(x-3) + (x-1)}{(x-1)(x-2)(x-3)} = \frac{2(x-2)}{(x-1)(x-2)(x-3)} = \frac{2}{(x-1)(x-3)}$$

$$\frac{2}{(x-1)(x-3)} = \frac{2}{3}, \text{ so } (x-1)(x-3) = 3$$

$$x^2 - 4x + 3 = 3, \text{ so } x(x-4) = 0$$

$$\text{Answer: } x = 0 \text{ or } x = 4$$

4 Solving by the quadratic formula

22. Solve for x : $\frac{1}{x+1} + \frac{2}{x+2} = \frac{4}{x+4}$, where $x \neq -1, -2, -4$. **CBSE 2016, 4 marks**

SOLUTION

$$\text{LHS} = \frac{(x+2) + 2(x+1)}{(x+1)(x+2)} = \frac{3x+4}{x^2+3x+2}$$

$$(3x+4)(x+4) = 4(x^2+3x+2), \text{ so } 3x^2 + 16x + 16 = 4x^2 + 12x + 8$$

$$x^2 - 4x - 8 = 0. \text{ Here } D = 16 + 32 = 48.$$

$$x = \frac{4 \pm \sqrt{48}}{2} = \frac{4 \pm 4\sqrt{3}}{2}$$

$$\text{Answer: } x = 2 + 2\sqrt{3} \text{ or } x = 2 - 2\sqrt{3}$$

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.

MathSetu